

WorkshopPlus: Azure AI Platform and Services

Azure AI Speech

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Objectives

After completing this Learning, you will be able to:

1

Understand AI Speech Technology

- ✓ Gain a comprehensive understanding of AI speech technology, including its history, development, and current applications.
- ✓ Learn about key algorithms and models used in the field.

2

Explore Practical Applications

- ✓ In different industries, demonstrating how it can enhance user experiences and improve operational efficiency.

3

Gain Hands-on Experience

- ✓ Engage in hands-on activities with AI speech tools.
- ✓ Gain insights into the practical aspects of implementing AI speech solutions.
- ✓ Understand potential challenges and solutions.

4

Practice Ethical AI

- ✓ Privacy, data security, and bias will be discussed.
- ✓ Emphasizing the need for responsible AI development and adherence to ethical guidelines



Prerequisites

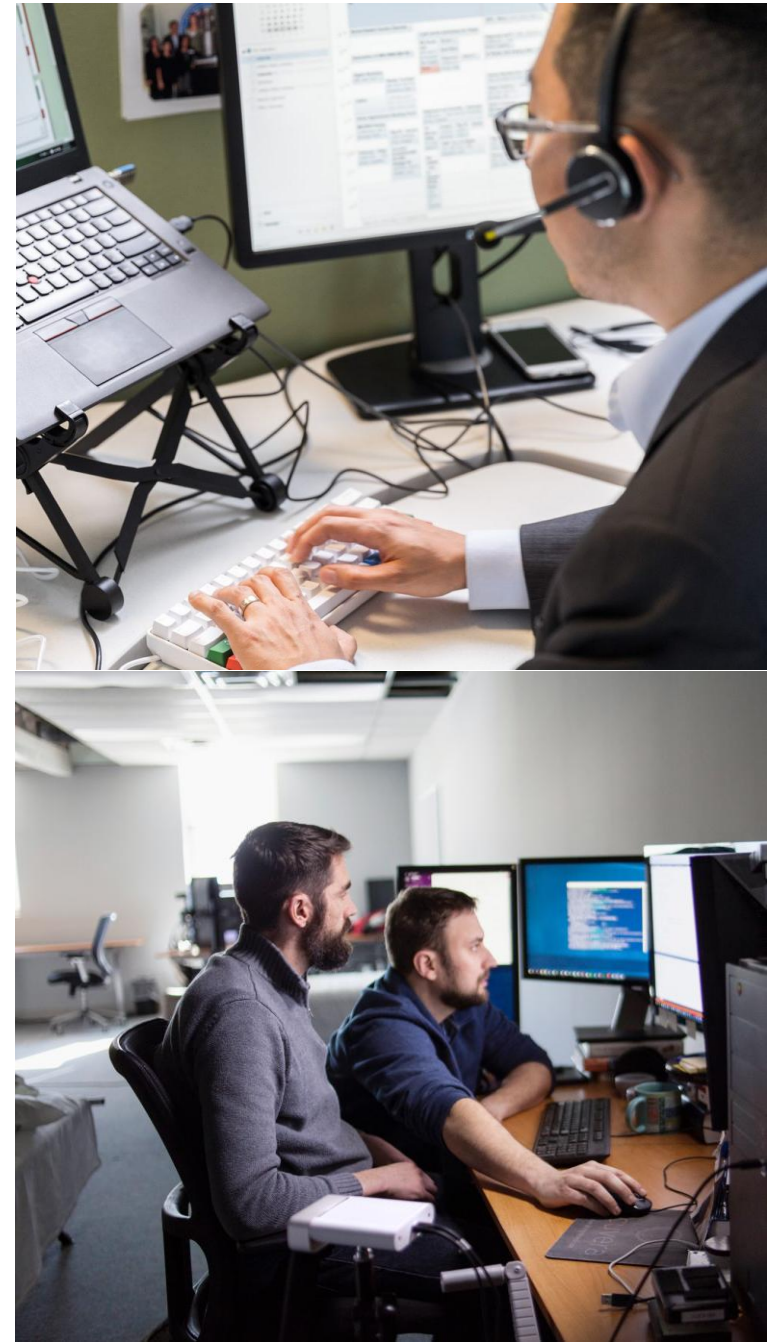
Basic Understanding of AI: Familiarity with fundamental concepts of artificial intelligence, including machine learning and neural networks, will help participants grasp the technical aspects of AI speech technology

Programming Knowledge: Basic programming skills, particularly in languages commonly used in AI development such as Python, will be useful for hands-on activities and understanding code examples

Interest in Speech Technology: A keen interest in speech recognition and synthesis technologies will enhance engagement and motivation during the workshop

Access to an Azure Subscription: Participants should have access to an Azure subscription to utilize Azure AI services and tools during the hands-on lab sessions

Technology Requirements: Ensure that participants have a laptop or computer with internet access, as well as the necessary software installed, such as Python and any required libraries or tools for the lab activities (more in the lab instructions)



Learning Units covered in this Module



Introduction to AI Speech Services



Speech to text



Text to speech



Speech capabilities by scenario



Introduction to AI Speech Services

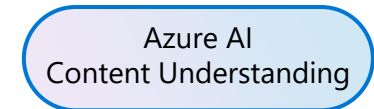
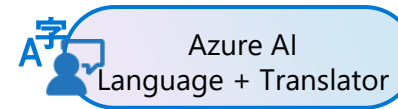
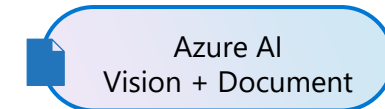
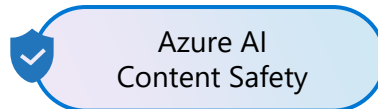
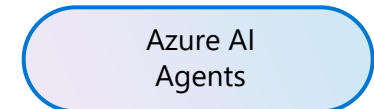
The Microsoft Azure AI Portfolio

Azure AI Foundry

The place to test, build and deploy AI solutions

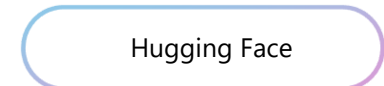
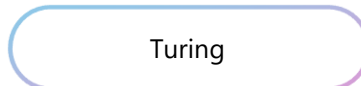
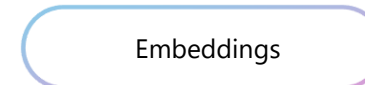
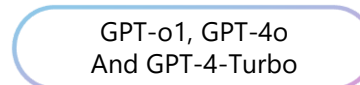
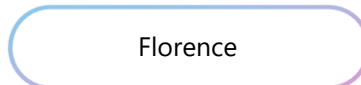
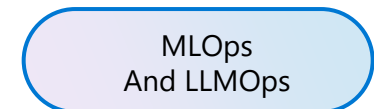
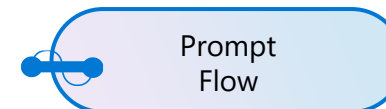
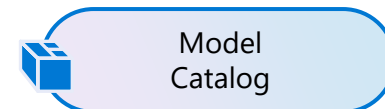
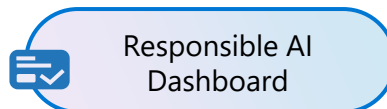
Azure AI Services

Pre-built models, APIs and SDKs to infuse into custom apps



Azure Machine Learning

Advanced tools for designing and fine-tuning specialized AI models



Azure AI Infrastructure

State-of-art supercomputing to power AI workloads

Build Intelligent Apps with Azure AI Services

Azure OpenAI Service

- Access to powerful AI models
- Scalable development
- Compliance & security
- Integration with other Azure Services

Azure AI Search

- AI enrichment & semantic ranking
- Generative AI content creation
- Vector search for data organization

Azure AI Speech

- Speech to text (including the Whisper model on Azure OpenAI Service)
- Text to speech
- Speech translation
- Speaker recognition

Azure AI Vision

- Image and face analysis
- Custom model training
- Face detection and recognition
- Document text extraction

Azure AI Content Safety

- AI-driven content moderation for enhanced safety
- Customize safety thresholds for diverse user types
- Detect and prevent Jailbreak Risk from XPIA attacks

Azure AI Document Intelligence

- Automated documentation generation
- Documentation quality analysis
- Interactive documentation experiences
- Natural language understanding for documentation

Azure AI Language

- Task-optimized AI models for text analytics
- Custom industry-specific AI for healthcare
- Custom, industry-specific models

Azure AI Translator

- Multilingual text and speech translation
- Synchronous and asynchronous translation request support
- Native translation of documents and manuals

Azure AI Speech Overview

SPEECH CAPABILITIES



Speech to Text
including the OpenAI Whisper Model



Text to Speech



Speech Translation



Speaker Recognition

AZURE TOOLS TO HELP YOU CUSTOMIZE AND INTEGRATE



Speech Studio



Ingestion Client



Batch Toolkit



Speech SDK

FLEXIBLE DEPLOYMENT OPTIONS



Cloud, On-Prem, On Edge

Azure AI Speech Speech to Text

Speech To Text Overview

Real-time Transcription

- Live transcription capabilities on your own audio without writing any code.

Fast Transcription

- Leveraging advanced speech recognition technology for rapid analysis and recognition.

Batch Transcription

- Transcribe a large amount of audio in storage and get results asynchronously

Speech Translation

- Translate speech into other languages of your choice with low latency.

Whisper from OpenAI

- Processes audio and converts into text in multiple languages whilst providing translation to English

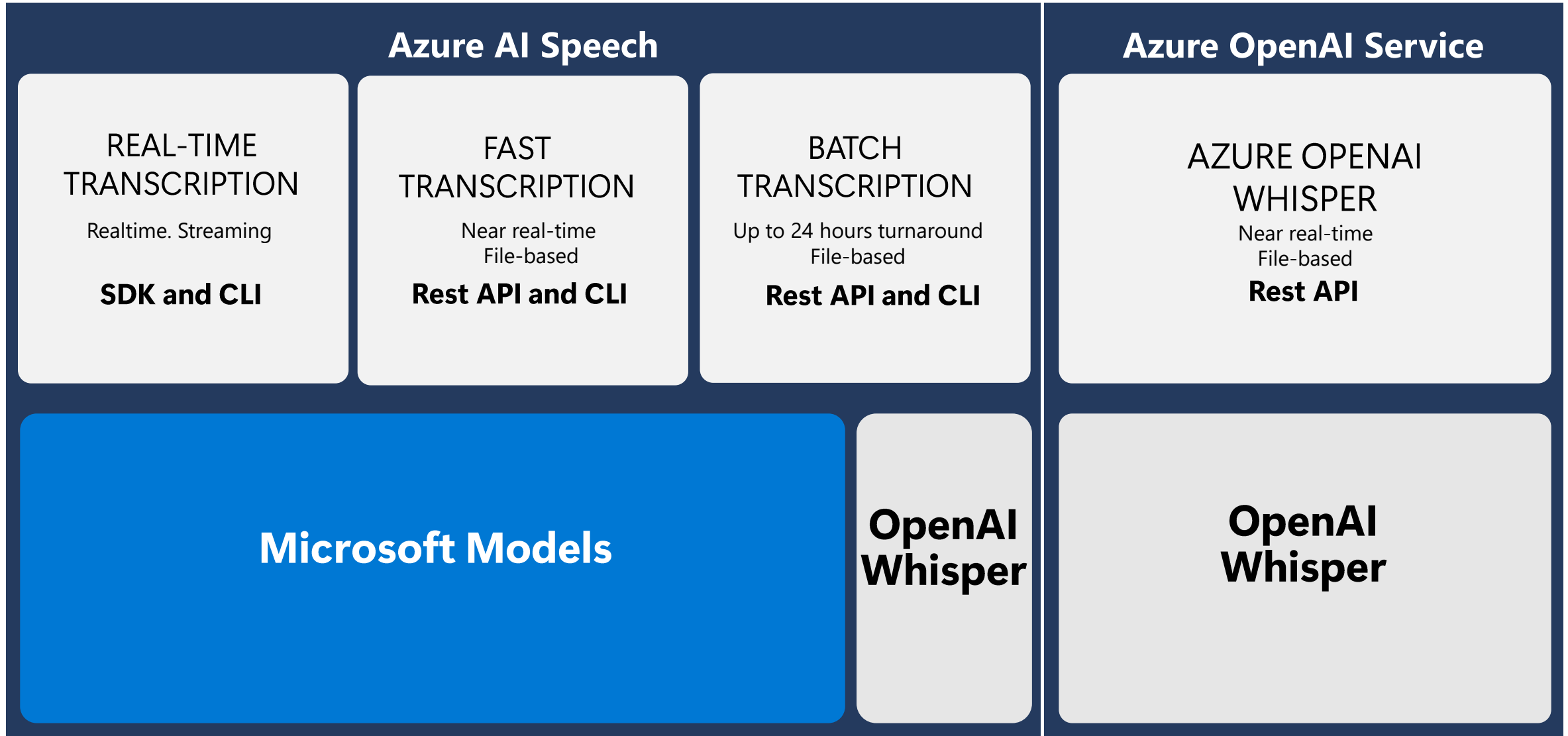
Pronunciation Assessment

- Get instant feedback on pronunciation accuracy and fluency by reading a script aloud

Custom Speech

- Customize speech to text with your data to adapt to specific speaking styles, vocabulary, and more.

3 Differentiated Speech To Text Offerings



Speech To Text Scenario



Transcription & Analytics

Contact center – post call analytics

Contact center – agent assist

Meeting transcription and insights

Audio/video transcription and insights

Dictation



Captioning & Subtitling

Online meeting caption & subtitle

In-person meeting caption & subtitle

Audio/video caption & subtitle

Speech To Text Real-time Transcription

Real-time Transcription

How it works?

- Transcribe from microphone / streaming audio, returning transcripts in real time with interim results.
- Realtime Factor ~1
- First response in a few hundred milliseconds

Good for?

- Transcription, captions or subtitles
- Diarization
- Pronunciation Assessment
- Dictation
- Voice Agent

Access?

- Speech SDK
- Speech CLI
- REST API



Speech To Text Fast Transcription

Fast Transcription

How it works?

- Transcribe one audio file per request, returning transcripts synchronously and quickly.
- Realtime Factor $\ll 1$
- Faster than real-time audio.
- Transcribe a 20 mins audio in less than 1 min.

Good for?

- Quick audio or video transcription and subtitles
- Video translation and immediate subtitles

Access?

- Speech CLI
- REST API



Speech To Text Batch Transcription

Batch Transcription

How it works?

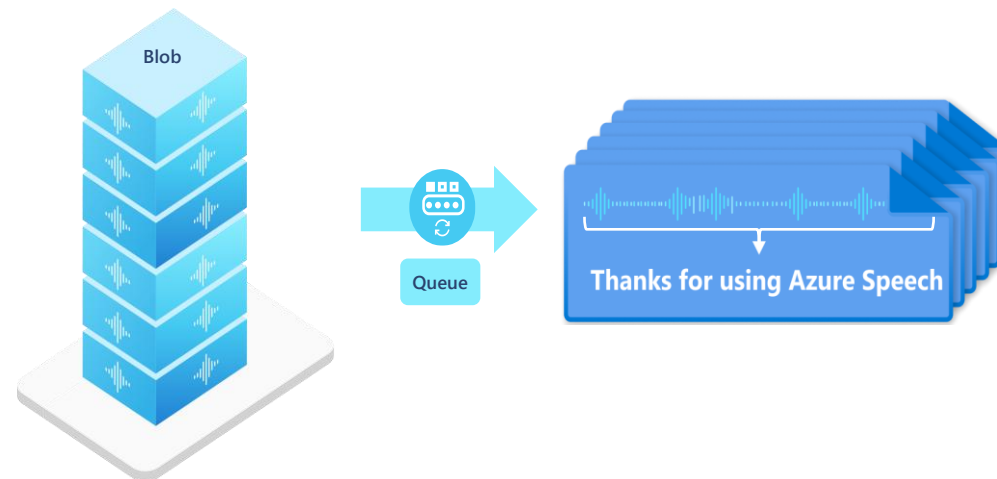
- Transcribe large amount of audio files in storage, returning transcripts asynchronously with low time urgency.
- Realtime Factor not guaranteed (Task queueing, low urgency)

Good for?

- Transcriptions, captions, or subtitles for prerecorded audio
- Contact center post-call analytics
- Diarization

Access?

- Speech CLI
- REST API



Speech To Text Whisper Model

Whisper Model

How it works?

- An encoder-decoder Transformer from OpenAI that processes 30 second chunks of audio and converts them into text in multiple languages.
- Realtime Factor not guaranteed (Task queueing, low urgency)

Good for?

- Quickly transcribing audio files one at a time
- Translate audio from other languages into English
- Prompt Tuning
- Transcribing files larger than 25MB (up to 1GB)
- Better readability

Access?

- REST API
 - Transcribe
 - Translate



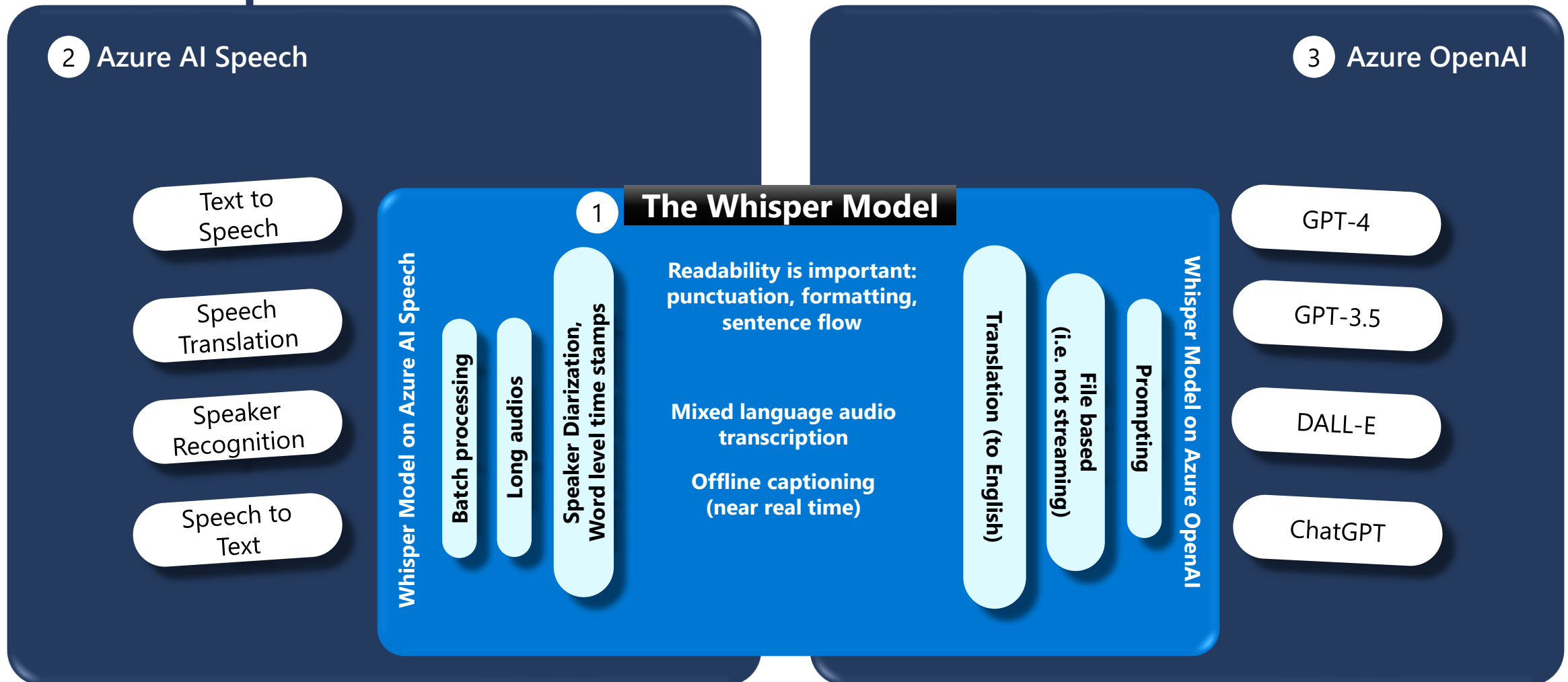
Using the Whisper Model in Azure

There are multiple ways customers can use the Whisper Model on Azure:

- **Via Azure AI Speech** – enables customers to use the Whisper Model with the Azure AI Speech batch transcription API
- **Via the Azure OpenAI Service** – mirrors the OpenAI API and its functionality
- **Via Azure Machine Learning** – through the Hugging Face model catalog. This is the Whisper open-source model that is part of the general open-source model catalog available in Azure ML.

The workshop is going to focus on the first two ways and the following slides are going to detail these further.

2 Ways to Access Whisper: Azure AI Speech and Azure OpenAI Service



The Whisper Model in Azure Supports Enterprise Readiness

Regional data processing

Developers control in which specific data center processing occurs for compliance with data protection regulations.

Reliability and Business Continuity

Azure AI Speech SLA: 99.9%. Business Continuity and Disaster Recovery through failover across Azure datacenters.

Security

Azure security plus support for VNet/private link, Role based access and Bring your own storage.

Privacy

Azure OpenAI Service and Azure AI Speech do not log or store customer data or use it to improve our general speech models.

Compliance

ISO, HIPAA, SOC and a broad range of other certifications.

Deployment flexibility

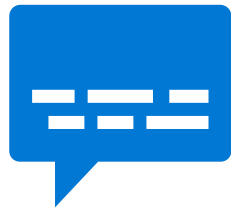
Azure AI Speech models: Cloud, containers and embedded speech recognition on devices.

Support

Support, up to 24x7 and <15min response time provided by Azure support.

Azure AI Speech – Whisper Model or Speech Model?

Within Azure AI Speech, one can choose whether they want to use the Whisper Model or the Azure AI Speech model. And the decision comes down to the scenario in question. Typical guidance is as follows:



Whisper Model

Captioning, where readability by humans is important

Mixed language audio.



Speech Model

Real-time streaming audio speech to text processing (e.g. live captioning or in agent assist)

Single language transcription and translation where Azure AI Speech models are more accurate (e.g. tail languages, telephony audio).

Azure AI Offers Transcription Technologies For Every Use Case

	<u>Real-time Transcription</u>	<u>Batch Transcription</u> (include Batch Whisper)	<u>Fast Transcription</u> (new)	<u>Whisper in Azure OpenAI Service</u>
Description	Transcribe live audio from microphone / stream , returning transcripts in real time with interim results .	Transcribe large amount of audio files in storage, returning transcripts asynchronously with low time-urgency .	Transcribe one audio file per request , returning transcripts synchronously and quickly .	A replication of OpenAI Whisper API hosted in Azure.
Input Form	Streaming audio	Audio file (many files via a blob storage URL)	Audio file (one audio file per request)	Audio file (one audio file per request)
Recommended Using Scenarios	- Online Meeting Caption/ Subtitle (translation) - In-person Meeting Caption/ Subtitle (translation) - Contact Center - Voice Agent - Contact Center – Agent Assist (streaming input with partial results) - Voice Assistant/Chatbot - Dictation - Voice Commanding	- Contact Center – Post Call Transcription and Analytics (lower urgency) - Meeting Transcription – Post Meeting Summarization and Analytics (lower urgency) - Recorded Audio/Video transcription (lower urgency)	- Quick Audio / Video edit - Contact Center – Agent Assist (file input) - Contact Center – Post Call Transcription and Analysis (higher urgency) - Meeting Transcription – Post Meeting Summarization and Analytics (higher urgency) - Recorded Audio/Video transcription (higher urgency) - Voice Mail - Video Dubbing	- Offline Audio/Video captioning/subtitling - Audio/Video Transcription - Speech translation to English only
Use if you need	- Streaming input with microphone; - Intermediate recognizing results - Improve quality with Phrase Lists - Improve quality with Custom Speech and deploy endpoints	- Bulk audio files in a storage in a request - Extreme large audio files with >2hrs audio length - Large selection of model versions - Audio formats/codec support in service side - Multiple audio channels - Best quality	- Simple config via one HTTP post - Transcribe much faster than real audio length - One audio file per request - Audio formats/codec support in service side - Good quality.	- Good readability and disfluency - Multilanguage recognition
Don't Use if you need	- Get transcript much faster than real audio time; - Audio formats/codec support in service side	- Streaming input; - Get transcript quickly and synchronously	- Streaming input; - Lexical format result (e.g., "one two three" instead of "123")	- Streaming input; - Lexical format result (e.g., "one two three" instead of "123") - Large audio files - Customization with your data
RTF (Realtime Factor) and Latency	~ 1 (realtime), first-byte latency response in a few hundred milliseconds	Not guaranteed due to task queueing (low time urgency)	<< 1 (e.g., 20 mins audio done in 1 min)	<< 1

Detailed Comparison Table: Supported Features

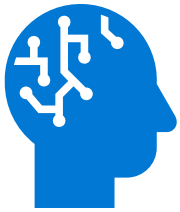
	<u>Real-time Transcription</u>	<u>Batch Transcription</u> (include Batch Whisper)	Fast Transcription (new)	Whisper in Azure OpenAI Service
Supported Language Locales	143+	143+	Top locales (~50 by Ignite 24)	~57
Display Format Output	Yes	Yes	Yes	Yes
Lexical Format Output	Yes	Yes	No	No
Intermediate / recognizing results?	Yes	No	No	No
Input format	WAV (PCM) (Others via integrating GStreamer)	WAV, MP3, OPUS/OGG, FLAC, WMA, AAC, ALAW in WAV container, MULAW in WAV container, AMR, WebM, M4A, SPEEX, etc	WAV, MP3, OPUS/OGG, FLAC, WMA, AAC, ALAW in WAV container, MULAW in WAV container, AMR, WebM, M4A, SPEEX, etc	mp3, mp4, mpeg, mpga, m4a, wav, webm
Audio File size limit	2GB	2GB or 4hrs audio	200MB or 2hrs audio	25MB
Word-level timestamp	Yes	Yes	Yes	Yes
Speaker Diarization	Yes	Yes	Yes	No
Custom Speech Model	Yes	Yes	Yes (planned)	No
Phrase List	Yes	No	No	No
Prompt tuning	No	No	No (planned)	Yes
Support "Auto Language ID"	Yes ("at-start" and "continuous" mode)	Yes ("single" and "continuous" mode)	Yes (planned, "single" mode)	Yes
Disfluency removal	Yes	Yes	Yes	Yes
Profanity mask	Yes	Yes	Yes	No
Confidence score	Yes	Yes	Yes	No

API Comparison

API Type	Description	Azure AI Speech models	Whisper model	Choice	
Realtime streaming	- Audio continuously streamed to service - Processed as audio comes in - Provides intermediate (word by word) results and final (utterance level) results with minimal lag	SDK (Websocket)	n/a		✓ Azure AI Speech Model
Synchronous single audio	- Single audio is in the request payload - Process as fast as possible - Limited to <15-30 minutes of audio depending on audio compression	REST Sync API	REST Sync API		✓ Whisper Model
Asynchronous bulk file processing	- Audio files in online storage - Process many files in parallel - Can handle very long files (hours of audio)	Batch Async API	Batch Async API		✓ Azure AI Speech Model

Whisper Azure AI Speech vs. Whisper Azure OpenAI Service

One can choose whether to use the Whisper Model via Azure OpenAI or via Azure AI Speech. And the decision comes down to the scenario in question. Typical guidance is as follows:



Via Azure OpenAI Service

Customers who are looking for a simple **API** to transcribe captured audio files

Customers who need to translate audio to English

Supported file formats: mp3, mp4, mpweg, mpga, m4a, wav and webm.

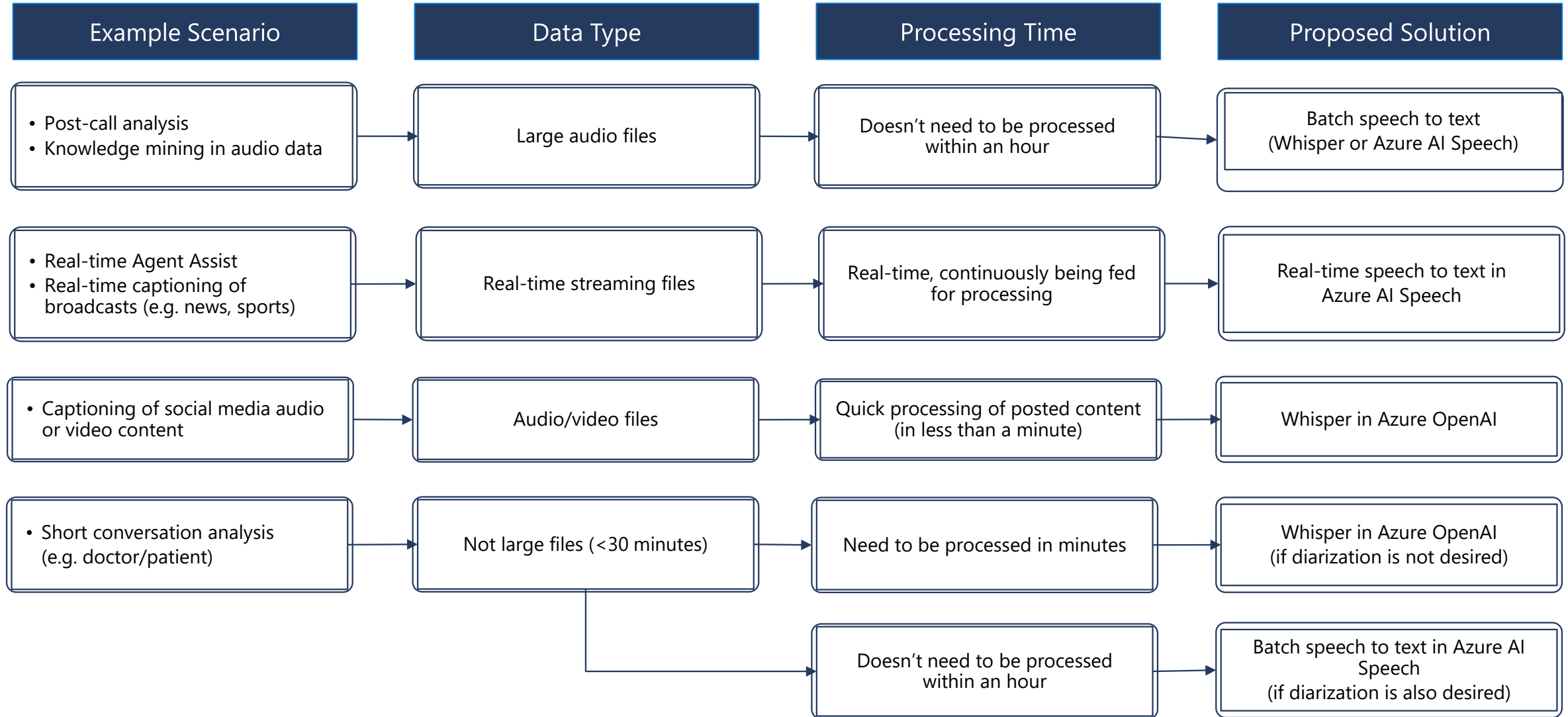


Via Azure AI Speech

Customers who need additional functionality:

- **Diarization**
- **Word-level** timestamps
- **Customization** of Whisper using Custom Speech (coming soon)
- **Transcribe** files larger than 25MB (up to 1GB)
- **Process** large batches of audio files
- Supported file formats: mp3, wav and ogg.

Scenario-to-Solution Mapping



Speech To Text

Pronunciation Assessment with Speech to Text

Pronunciation Assessment with Speech to Text

How it works?

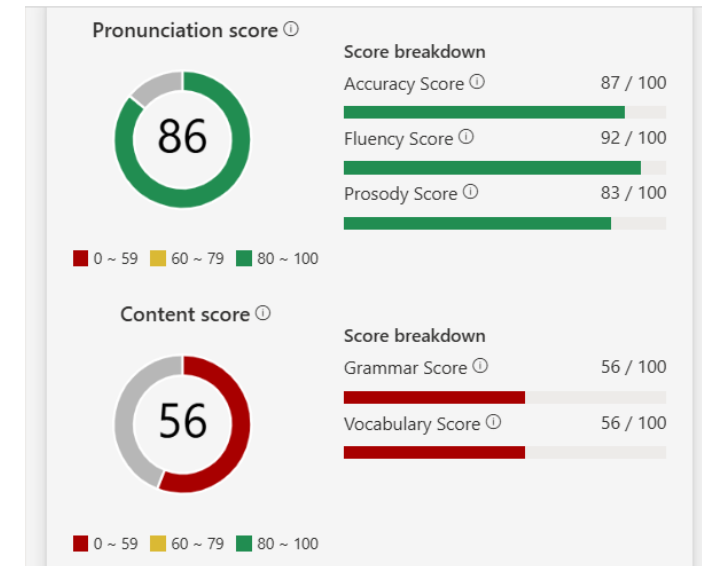
- Evaluates pronunciation, fluency, and accuracy in speech.
- Provides detailed pronunciation feedback at phoneme, word, and text levels
- Supports multiple languages, including English, Chinese, French, German, Japanese, and Spanish.

Good for?

- Enhances language learning applications, education tools, and corporate training.
- Designed to help language learners and educators improve spoken language skills.

Access?

- Speech SDK



Comprehensive Pronunciation Evaluation for Reading (1 of 2)

Accuracy Score

Pronunciation accuracy of the speech. Accuracy indicates how closely the phonemes match a native speaker's pronunciation.

Fluency Score

Fluency indicates how closely the speech matches a native speaker's use of silent breaks between words.

Completeness Score

Completeness of the speech, calculated by the ratio of pronounced words to the input reference text.

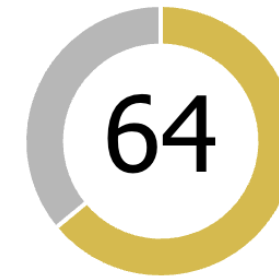
Prosody Score

Prosody indicates how natural the given speech is, including stress, intonation, speaking speed, and rhythm.

Pronunciation Score

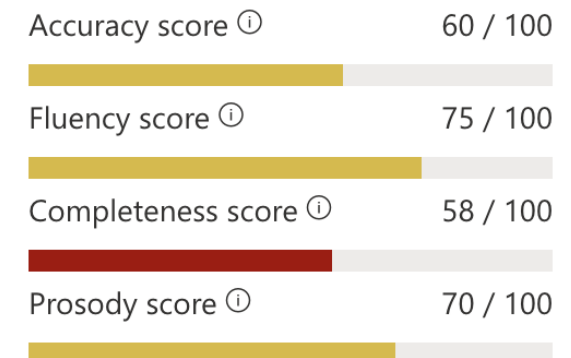
Overall score of the pronunciation quality of the given speech. Calculated from Accuracy Score, Fluency Score, Completeness Score, and Prosody Score.

Pronunciation score ⓘ



■ 0 ~ 59 ■ 60 ~ 79 ■ 80 ~ 100

Score breakdown



Comprehensive Pronunciation Evaluation for Reading (2 of 2)

Error Type

This value indicates the error type on word level compared to the reference text. Possible values are:

- None for no error
- **Omission:** The words that are provided in the script but are not spoken.
- **Insertion:** The words that are not in the script but are detected in the recording.
- **Mispronunciation:** The words that are spoken incorrectly
- **Unexpected Break:** Improperly paused in between words within same sentence
- **Missing Break:** The words that are not in the script but are detected in the recording.
- **Monotone:** The words are being read in a flat and unexciting tone, without any rhythm or expression.



it  took me a long time to  learn

 where  he  came  from.

Errors

1	Mispronunciations ⓘ	☒ On
4	Omissions ⓘ	☒ On
0	Insertions ⓘ	☒ On
1	Unexpected break ⓘ	☒ On
0	Missing break ⓘ	☒ On
0	Monotone ⓘ	☒ On

Comprehensive Pronunciation Evaluation for Speaking (1 of 2)

Vocabulary Score

Proficiency in lexical usage. It evaluates the speaker's effective usage of words and their appropriateness within the given context to express ideas accurately, and the level of lexical complexity.

Grammar Score

Correctness in using grammar and variety of sentence patterns. Lexical accuracy, grammatical accuracy, and diversity of sentence structures jointly elevate grammatical errors.

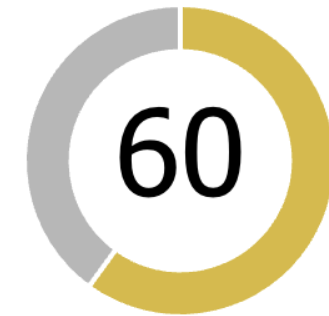
Topic Score

Level of understanding and engagement with the topic, which provides insights into the speaker's ability to express their thoughts and ideas effectively and the ability to engage with the topic.

Content Score

Calculated by Vocabulary Score, Grammar Score and Topic Score

Content score ⓘ



■ 0 ~ 59 ■ 60 ~ 79 ■ 80 ~ 100

Score breakdown

Vocabulary score ⓘ 54 / 100



Grammar score ⓘ 50 / 100



Topic score ⓘ 76 / 100



Comprehensive Pronunciation Evaluation for Speaking (2 of 2)

Error Type

This value indicates the error type when a word is badly pronounced, improperly inserted with a break, or missing a break at punctuation. Possible values are:

- None for no error
- **Mispronunciation:** The words that are spoken incorrectly
- **Unexpected Break:** Improperly paused in between words within same sentence
- **Missing Break:** The words that are not in the script but are detected in the recording.
- **Monotone:** The words are being read in a flat and unexciting tone, without any rhythm or expression.



[Sample response]

today was a burst of adventures. i am  wake up bright and early to help my dad make pancakes.  later we set them  on a nature hike.  we spotted some **fascinating** birds.  after that  i joined my friends for a lively virtual dance class  and we had an absolute blast,  grooving and laughing together.

Errors

1	Mispronunciations ⓘ	☒ On
4	Unexpected break ⓘ	☒ On
4	Missing break ⓘ	☒ On
0	Monotone ⓘ	☒ On

Speech To Text

Speech Translation

Pronunciation Assessment with Speech to Text

How it works?

- Ability to take in an input audio stream in your specified source language and have it translated and outputted as text in your specified target language.
- Also offers read aloud the translated text using pretrained voices, allowing for a natural output of the input speech

Good for?

- Travel Interpreter
- Business Meeting
- Multi-lingual translation

Access?

- Speech SDK
- Speech CLI
- REST API

File name: 20250201_143031.wav Language: Thai (Thailand)
Target language: English Voice name: en-US (en-US-JennyNeural)

00:00  00:08 

Translated text Original text JSON 

```
2
3  "privTranslationPhrase": {
4    "Duration": 48000000,
5    "Offset": 6300000,
6    "RecognitionStatus": 0,
7    "Text": "สวัสดีค่ะ วันนี้จะมาเล่าให้ฟังเกี่ยวกับการแปลภาษา",
8    "Confidence": 0,
9    "Translation": {
10     "TranslationStatus": 0,
11     "Translations": [
12       {
13         "Language": "en",
14         "Text": "Hello, today I will tell you about language translation."
15       }
16     ]
17   }
18 }
19
20
```

Speech Translation API

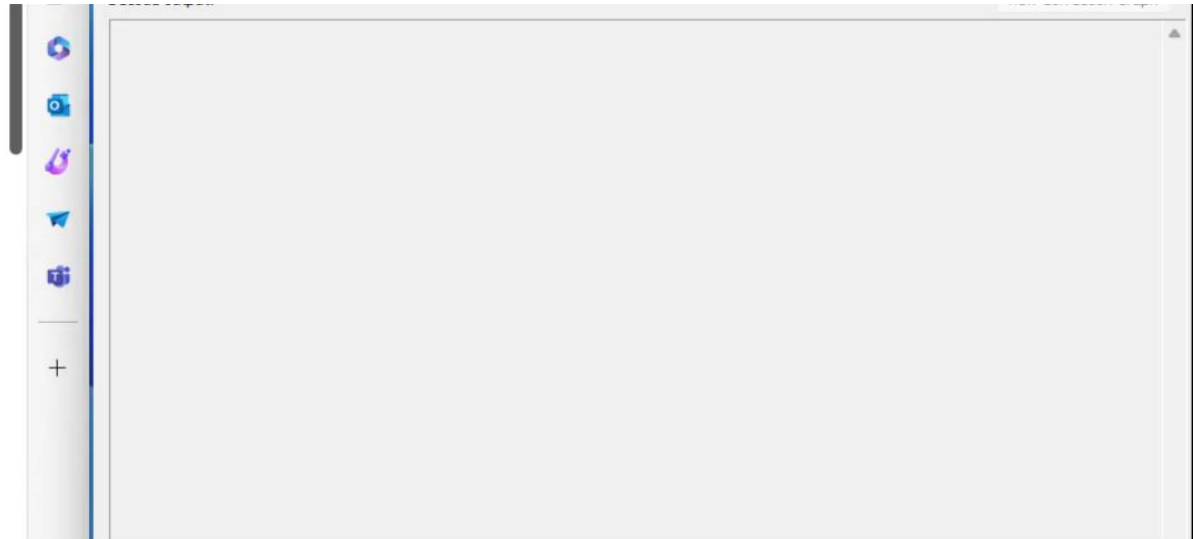
Translate audio streams into text & audio in target languages

Realtime

- **Multiple language detection:** model will detect multiple languages among the supported languages in the same audio stream
- **Automatic language detection:** no need to specify input languages – model will detect them automatically
- **Integrated custom translation** to adapt model to your domain-specific vocabulary

Powering business scenarios such as:

- **Live translated captions for videos**
 - Movies, TV shows & documentaries
 - Social media livestreams & short-form videos
 - Corporate education & training videos
- **Contact center** live translated call scripts
- **Travel interpreter** apps



Speech To Text

Custom Speech

Custom Speech

How it works?

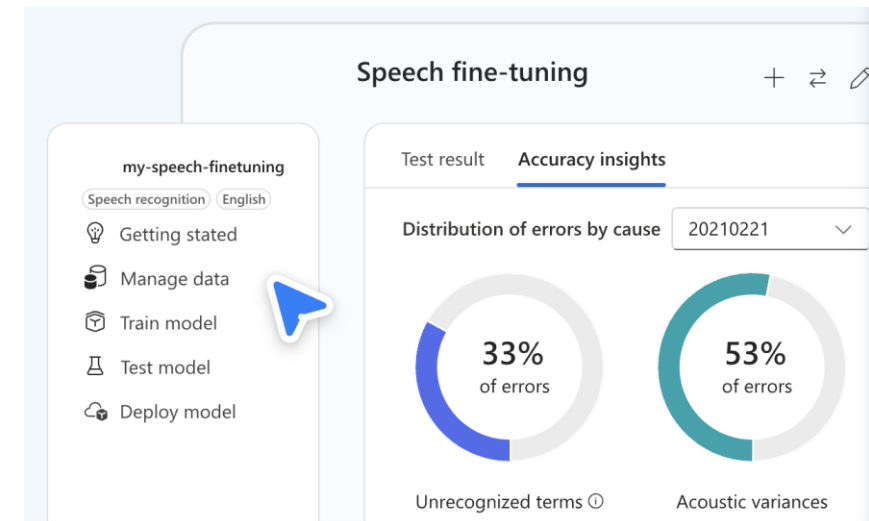
- With custom speech, you can upload your own data, test and train a custom model, compare accuracy between models, and deploy a model to a custom endpoint.
- Also offers read aloud the translated text using pretrained voices, allowing for a natural output of the input speech

Good for?

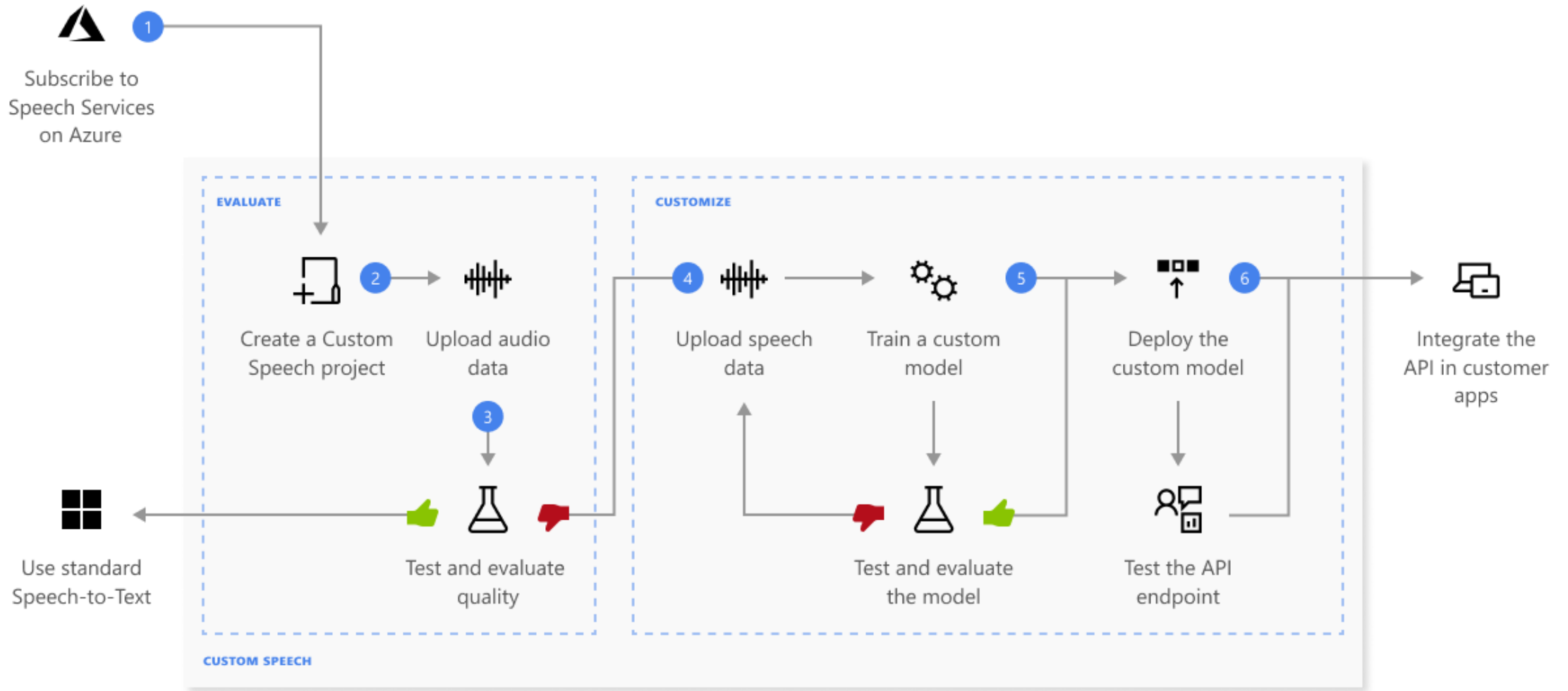
- Any use case where speech to text accuracy in your application scenarios remains low, consider customizing the model for your acoustic and linguistic variations i.e. Biology, Media Entities.

Access?

- Speech Studio
- Speech CLI
- REST API
- Foundry Portal



Custom Speech Architecture



Text-Based Custom Speech Improvements

Terminology

- Improve recognition accuracy on industry-specific domain.
- Dinosaur names:
Colepiocephale
Piatnitzkysaurus
Yamaceratops
Bambiraptor
Linhenykus
Jeholopterus

Pronunciation

- Improve recognition of a word or term that has nonstandard pronunciation
- Spoken format:
 - "3CPO" as "three c p o"
 - "IEEE" as "i triple e"

Sentence Structure

- Recognize a particular pattern in utterances
- "I have a question about **@product**," where **@product** is a list of possible products.
- @product = apple, orange, banana, kiwi, strawberry, watermelon

Audio-Based Custom Speech Improvements

Acoustic variance

- Improve recognition accuracy on speaking styles, accents, or specific background noises.
- A coffee franchise expects all drive-through ordering to be done outside with a specific microphone model, audio recordings in those condition were collected to improve recognition for acoustic variance.

Output Format

- Create output text format rules such as reformulation, profanity filter, inverse-text-normalization pattern (ITN).
- **"fifty-three s" to "53-S"**
- **"Mask profanity" to "Mask *****"**
- **"t9324523" to "T-932-4523"**

AZURE SPEECH

Customization Demo



Azure AI Speech

Text to Speech

Text to Speech Overview

Voice Gallery

- Browse expressive voices with humanlike speech to find the perfect speaker for your project

Custom Neural Voice

- Use your own audio to create a distinct, one-of-a-kind voice for your text to speech apps.

Personal Voice

- Create AI easily from human voice sample, support 100+ languages

Text-to-Speech Avatar

- Bring text to life with natural sounding voices and photorealistic talking avatars

Audio Content Creation

- Craft nuanced speech by adjusting the speaking style, pacing and pronunciation



Text to Speech Voice Gallery

Custom Speech

How it works?

- Build apps and services that speak naturally, choosing from 600 voices across 154 languages.
- Understands the underlying sentiment
- Automatically adjusts the speaking tone to match the emotion conveyed

Good for?

- Any use case where you bring your scenarios to life with highly expressive and humanlike neural voices.

Access?

- Speech SDK
- REST API
- AI Foundry Portal



de-DE-Seraphina



en-US-Andrew



en-US-Ava



en-US-Davis



zh-CN-Xiaochen

OpenAI TTS voice demo

Voice gallery

Build apps and services that speak naturally, choosing from 497 voices across 149 languages and variants. Bring your scenarios to life with highly expressive and humanlike neural voices.

[← Back](#) [View documentation](#) [View on GitHub](#) [Use SDK reference](#) [Use REST API](#)

Explore voices

Azure AI Services *

aaaaapr22507732335667

Language

English (United States)

Sort by

Default

☐ Show only favourite

44 voices

Filter

Open AI voices are only available in two service locations: Sweden Central and North Central US. Switch to an AI services resource in the supported location to view those voices.

**Ava Multilingual**
Female
English (United States) +90

A bright, engaging voice with a beautiful tone that's perfect for delivering search results and capturing users' attention.

Pleasant Chat +3

FEATURED

**Andrew Multilingual**
Male
English (United States) +90

A warm, engaging voice that sounds like someone you want to know, perfect for building a connection with listeners.

Confident Chat +3

FEATURED

**Emma Multilingual**
Female
English (United States) +90

A friendly, sincere voice with a light-hearted and pleasant tone that's ideal for education and explanations.

Cheerful E-learning +3

POPULAR

**Brian Multilingual**
Male
English (United States) +90

A youthful, cheerful, and versatile voice that can handle any task you throw its way, well-suited to a wide variety of contexts.

Sincere Podcast +3

POPULAR

**Ava**
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English (United States)

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**Brian**
Male
English (United States)

A youthful, cheerful, and versatile voice that can handle any task you throw its way, well-suited to a wide variety of contexts.

Sincere Podcast +3

**Jenny**
Female
English (United States)

15 styles
A youthful voice with a wide range of expressions, perfect for customer service and keeping users satisfied.

Sincere News +3

**Guy**
Male
English (United States)

12 styles

**Aria**
Female
English (United States)

17 styles

**Davis**
Male
English (United States)

12 styles

Voice details Sample code

**Ava Multilingual**
Female
English (United States) +90

A bright, engaging voice with a beautiful tone that's perfect for delivering search results and capturing users' attention.

Speaking styles: 1

Default

Languages: 91

Afrikaans (South Africa) Albanian (Albania) Amharic (Ethiopia) Arabic (Egypt) Arabic (Saudi Arabia) Armenian (Armenia) Azerbaijani (Azerbaijan) Bangla (Bangladesh) Basque (Basque) Bengali (India) Bosnian (Bosnia and Herzegovina) Bulgarian (Bulgaria) Burmese (Myanmar) Catalan (Spain)

Open in Playground

600+ prebuilt voices available out of box



Demonstration

zh-CN- XiaoxiaoMultiligualNeural (Female) (女)	Conversation 对话	但我现在对这个职业的热爱还是非常的，呵呵,非常的，嗯,怎么说呢？日月可鉴的，哈哈，嗯还是希望可以把这个职业做下去或者做这个声音相关领域的工作，嗯，就是把自己的优势发挥的大一点，尽可能能用到自己擅长的东西，而不是说为了工作，为了挣钱而工作。 ▶ 0:22 / 0:22 ————— 🔊 ⋮
	Podcast 播客	我们需要加的是生抽、老抽、料酒、白糖还有一点点的醋、盐，然后把它翻炒均匀就可以了。接下来就是收汁的阶段了哈，我们加入适量的水淀粉翻炒到这个鸡丁上色，而且汤汁呢，稍稍已经比较浓稠，啊不会轻易的滑落。 ▶ 0:00 / 0:19 ————— 🔊 ⋮

Different Languages and Variations

Multi-style

VOICE -DOMAINS	SAMPLE
Ava – Chat (Interjections)	
Ava - Poem	
Andrew - Chat (Interjections)	
Andrew – Sports	
Emma - Chat (Interjections)	
Emma – e-learning	
Brian – Chat (Interjections)	
Brian – Advertisement	

Multi-lingual

VOICE	SAMPLE LOCALE(S)	SAMPLE
Ava	English – French – Italian	
Andrew	English – German – Spanish	
Emma	English – Portuguese – Chinese	
Brian	English – Japanese – Korean	

More Example Voices - Super realistic voices for conversations



De-DE
Seraphina



En-US
Emma



En-US
Andrew



En-US
Brian



Es-ES
Ximena



Fr-CA
Thierry



Fr-FR
Vivienne



It-IT
Giuseppe



Ko-KR
Hyunsu



Pt-BR
Thalita



Zh-CN
Yunjie

Conversational voices that can support diverse domains for Copilot, and speak multiple languages

Text to Speech

Custom Neural Voice

Custom Neural Voice

How it works?

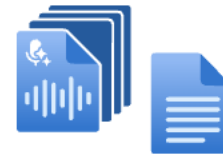
- Build a highly natural-sounding voice for your brand or characters by providing human speech samples as training data.
- Create a one-of-a-kind, customized, synthetic voice for your applications.

Good for?

- Brand identity
- Natural interaction
- Custom persona

Access?

- Speech SDK
- REST API
- AI Foundry Portal



Voice recordings and transcripts



Train by Custom Neural Voice



Synthetic voice for your brand

[Azure AI Limited-Access Feature](#)

Demo Custom Neural Voice

Press microphone to start recording then speak following script clearly.

Maybe it's going home where you grew up. 



Try again

 00:00

00:04s

Clearness ⓘ

Excellent

Pronunciation ⓘ

Excellent

Volume ⓘ

Excellent

Omissions ⓘ

0

Mispronunciations ⓘ

0

Insertions ⓘ

0

 **20 samples required** to train or record up to 50

20 recorded

Record next

Recorded samples					
Sample ID			Script	Recording quality	Last updated ↓
000000020			As evening approaches, tennis, volleyball, and softball get going.		1/15/2022 3:01 AM
000000019			What kind of unforeseen adventures has it sparked?		1/15/2022 3:00 AM
000000018			I am in a hurry right now!		1/15/2022 3:00 AM
000000017			Maybe it's going home where you grew up.		1/15/2022 3:00 AM
000000016			If you stop talking about this right now, we can just pretend this whole thing never happened.		1/15/2022 3:00 AM
000000015			Have you met them before?		1/15/2022 2:59 AM



Demo Custom Neural Voice – Crosslingual



Human
recording
(300)

You don't need to
provide additional
language training data
to gain multilingual
skills through language
transfer learning



en-US



en-GB



zh-CN



fr-FR



de-DE



pt-BR



ko-KR



ja-JP

Train a new model

☒ Select training method

☐ Choose data

☐ Choose test script

☐ Name your voice

☐ Review and train

Select the training method for your model

Neural

Create a highly realistic voice in the same language of your training data. Start with 300 utterances.

Neural - cross lingual

Create a voice that speaks a different language from your training data. Start with 300 utterances.

Neural - multi style

Create a voice that speaks in multiple styles. Start with 300 utterances in a default style and transfer it to express different emotions without training data required. You can further create custom styles for the same voice if you have additional style data available.

Version ⓘ

V5.2023.07

Version description: Model updated for higher voice naturalness, together with rich pitch contour tuning. 25-50 compute hours estimated for each training.

Target language

Select the language that you would like your synthetic voice to speak

Next

Cancel

Custom Neural Voice – Multi-Style



Custom neural
voice without
emotional style



Angry



Terrified



Excited



Whispering



Sad

Train a new model

- Select training method
- Select preset speaking styles
- Add custom speaking styles
- Choose a speaker file
- Choose test script
- Name your voice
- Review and train

Select the training method for your model

Neural
Create a highly realistic voice in the same language of your training data. Start with 300 utterances.

Neural - cross lingual
Create a voice that speaks a different language from your training data. Start with 300 utterances.

Neural - multi style
Create a voice that speaks in multiple styles. Start with 300 utterances in a default style and transfer it to express different emotions without training data required. You can further create custom styles for the same voice if you have additional style data available.

Version
V1.2022.12

Version description: Basic version to adapt the voice to speak with multiple emotional styles, or create your own speaking style with custom style training data. 90-95 computer hours estimated for each training. Style degree tuning supported.

[Apply if you do not have the access to Custom Neural Voice.](#)

Next

Cancel

Select preset speaking styles

Choose the styles you'd like to train. Listen to the sample audios below to understand how each style sounds.

- ☒ Select all
- ☐ Default

☒ Angry

☒ Excited

☒ Cheerful

☒ Friendly

☒ Hopeful

☒ Sad

☒ Shouting

☒ Terrified

☒ Unfriendly

☒ Whispering

Custom Neural Voice Lite vs Professional

Items	Lite (Preview)	Pro
Target scenarios	Demonstration or evaluation	Professional scenarios like brand and character voices for chat bots, or audio content reading.
Training data	Record online using Speech Studio	Bring your own data. Recording in a professional studio is recommended.
Scripts for recording	Provided in Speech Studio	Use your own scripts that match the use case scenario. Microsoft provides example scripts for reference.
Required data size	20-50 utterances	300-2000 utterances
Training time	Less than one compute hour	Approximately 20-40 compute hours
Voice quality	Moderate quality	High quality
Availability	Anyone can record samples online and train a model for demo and evaluation purpose. Full access to custom neural voice is required if you want to deploy the CNV lite model for business use.	Data upload isn't restricted, but you can only train and deploy a CNV professional model after access is approved. CNV professional access is limited based on eligibility and usage criteria. Request access on the intake form .
Pricing	Per unit prices apply equally for both the CNV lite and CNV professional projects. Check the pricing details here .	Per unit prices apply equally for both the CNV lite and CNV professional projects. Check the pricing details here .

Text to Speech Personal Voice

Personal Voice

- Create a personalized AI voice in 10 seconds, providing your users with true voice digital experience across 100 languages.

Cognitive Services | Speech Studio

Speech Studio > Personal Voice

Personal Voice

Create an AI voice easily from a human voice sample, providing your users with a personalized voice experience across 100 languages.

Platforms
Cloud

Make each voice experience personalized

Voice assistant
Create a more personalized voice assistant experience. Instead of relying on a generic voice, users can now use their own voice for a truly unique experience.

Gaming
Enable an immersive experience for gamers, allowing them to fully embody their characters with their own unique voices.

Language dubbing
Expand your global reach and dub your content into 100 languages with the speakers' native voice for a seamless and enjoyable experience for your viewers, no matter where they are located.

Entertainment media
Create easy-to-use personal voices for stories, audio books, podcasts, videos, and more, making the content more relatable and immersive than ever before.

Accessibility
Enable people to build their own voice and use it for natural communications.

Speech translation
Allow your meeting participants to be heard in various languages with their true voice.

View documentation

View samples on GitHub

Try the samples

Explore samples of human voice prompts and their replicas for different scenarios and across multiple languages.

Scenario-specific

Emotional

Multi-lingual

Voice assistant

Prompt

Generated

Broadcast

Prompt

Generated

Conversation

Prompt

Generated

Story

Prompt

Generated

Try your own voice

Record your voice and create an AI replica in seconds. With each standard Speech resource, you can try up to five voices for free. If you want to use this service for your business, please apply for access through [this link](#).

Use with your resource

☒ I acknowledge that this application uses the resource AZ_Test and will incur usage to my account. [Choose a different resource](#)

Voice list

+ New voice

Winnie May_202306281737

Detail

Winnie May_202306281737

Voice talent

Winnie May

Verbal statement

Source language

English (United States)

Generate audios using sample scripts

Select an output language to switch scripts across languages. [Check all the languages supported](#)

Output language

English (United States)

Base model

PhoenixV2

Can you say that last part again?

Generate

This is the story of how she got there, where she is going, and how she wants the world to change. T...

Generate

Alright, can you tell me your message again?

Generate

You don't have any reminders for Tuesday evening.

Generate

There's nothing scheduled for 8 AM.

Generate

Personal Voice – Feature Highlights



Highly Personalized Interactions

Realistic voices and human-like expressions. This creates a more personal and engaging experience.



Extremely small samples

Preparing training samples for creating an AI voice could be difficult or costly. With personal voice, users can create a voice that just sound like them with a voice sample, as short as 10 seconds.



Express

Personal voice greatly reduced waiting time, allowing users to create a voice in seconds.



Multilingual and global reach

With an audio prompt in one language, the voice created can be used in 100 languages & variances, reaching a global audience.



Cost effective

Minimize the time to market with easy-to-use APIs, low resource required.

Prompt



TTS



Text to Speech

TTS Avatar

Text to Speech Avatar

- With text to speech avatar's advanced neural network models, the feature empowers users to deliver life-like and high-quality synthetic talking avatar videos for various applications while adhering to [responsible AI practices](#).

[Azure AI Limited-Access Feature](#)

Speech Studio > Talking Avatar

Text to speech Avatar

Turn your text into a video. You can select an AI-generated avatar and a realistic voice to bring your content to life.

[View documentation](#) [View sample code](#)

[Try it out](#) [View sample code](#) [Next steps](#)

Try it out

Use with your resource

☒ I acknowledge that this application uses the resource ACC-LE-SEA and will incur usage of my account. [Choose a different resource](#)



Avatar Background Video generated


Harry
business


Harry
casual


Harry
youthful


Jeff
business


Jeff
formal


Lisa
casual-sitting

[Generate video](#)

[Play audio](#) [Stop playing](#) [Insert break](#) [Insert gesture](#) [Speaking speed](#) [SSML mode](#)

Language English (United States) Voice Jenny Multilingual

Hello.

Microsoft Confidential

Prebuilt Avatar

- **2D Photorealistic digital humans:** The future of virtual interactions
- **Limited data, maximum output:** Our avatar models are trained with minimal video recordings
- **Text-triggered responses:** Create scripts with natural language
- **Multilingual capabilities:** High-quality neural TTS and custom neural voices for global reach
- **Batch or real-time streaming**

Prebuilt avatar

Use our prebuilt avatar in content generation or interactive scenarios



Customer service
Live chatbot

Training video
Conference speech

Custom Avatar

- One-of-a-kind synthetic talking avatar
- Train from your sample consent video recordings
- Gestures added to live chats
- Even more realistic with Custom Neural Voice
- Support live chat and video generation

Custom avatar

Build your own branded avatar with us by providing **10 mins** video recording

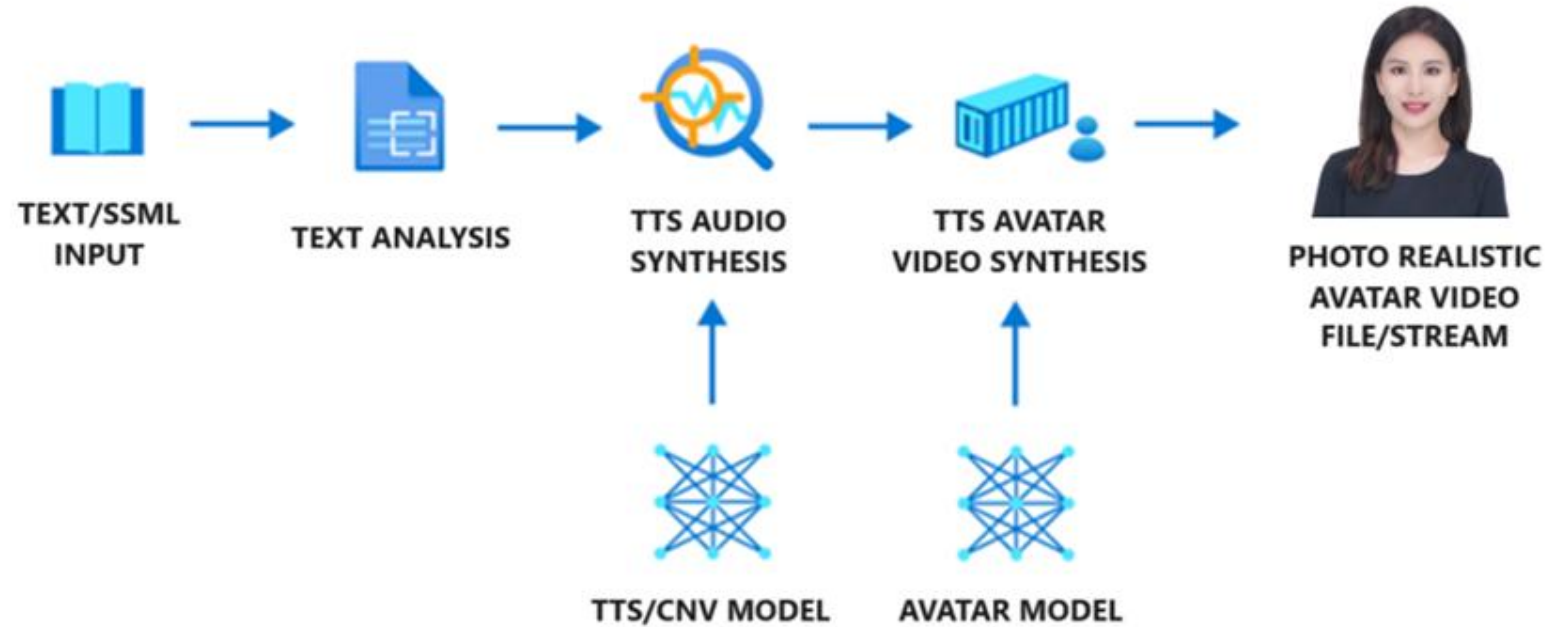


CEO/leader avatar
Promote brand

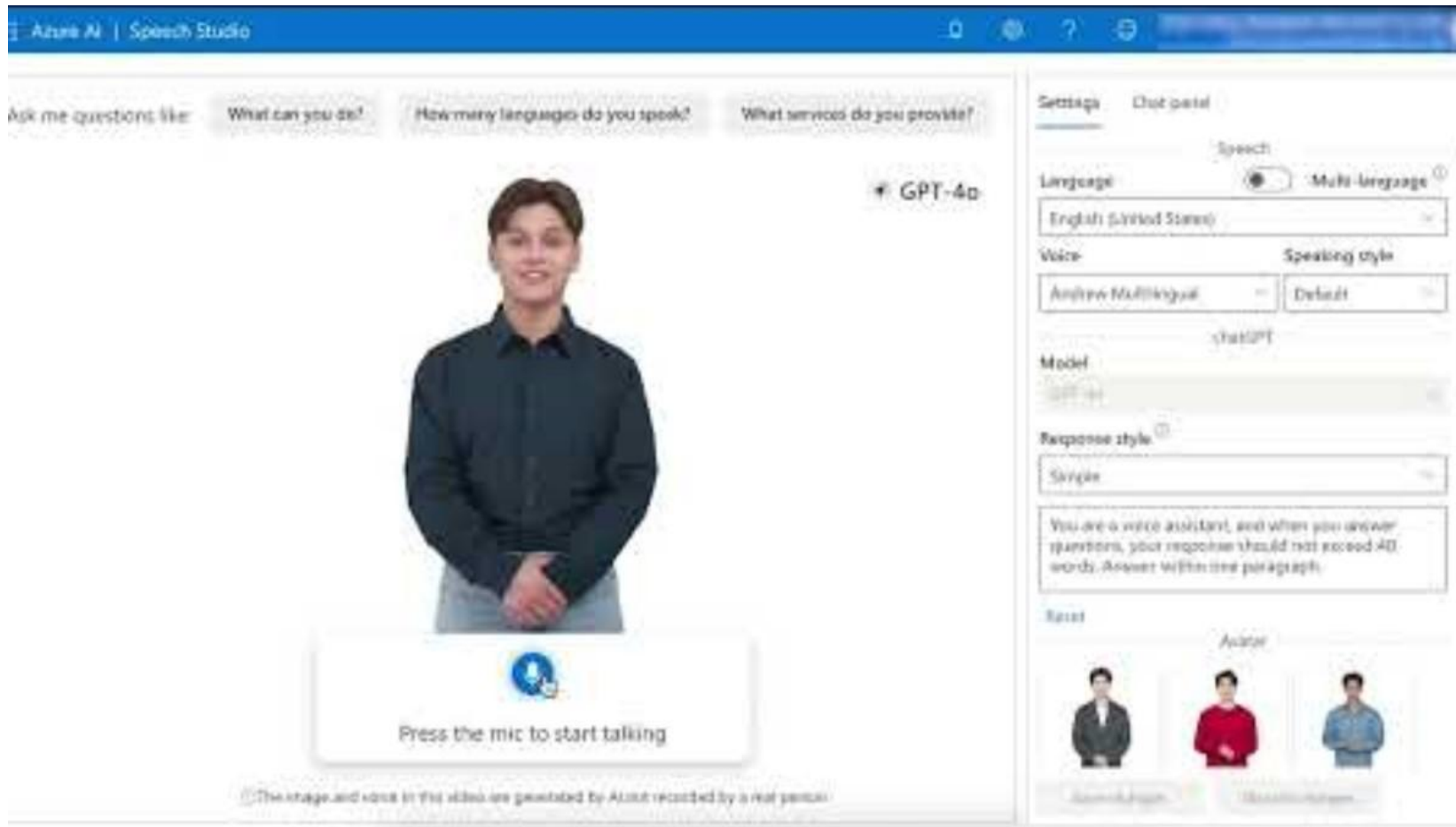
Live streaming
Ads

[Azure AI Limited-Access Feature](#)

Architecture



Demonstration



Deploy the Solution with GitHub Repository



How to Get Started:

1. Clone the repository to your local machine.
2. Follow the README guide to setup the local web application.
3. Deploy the web application and experience the solution in action.

Quick Access: [GitHub - aadrikasingh/Azure-Text-To-Speech-Avatar](https://github.com/aadrikasingh/Azure-Text-To-Speech-Avatar)

Text to Speech Audio Content Creation Tool

Audio Content Creation Tool

How it works?

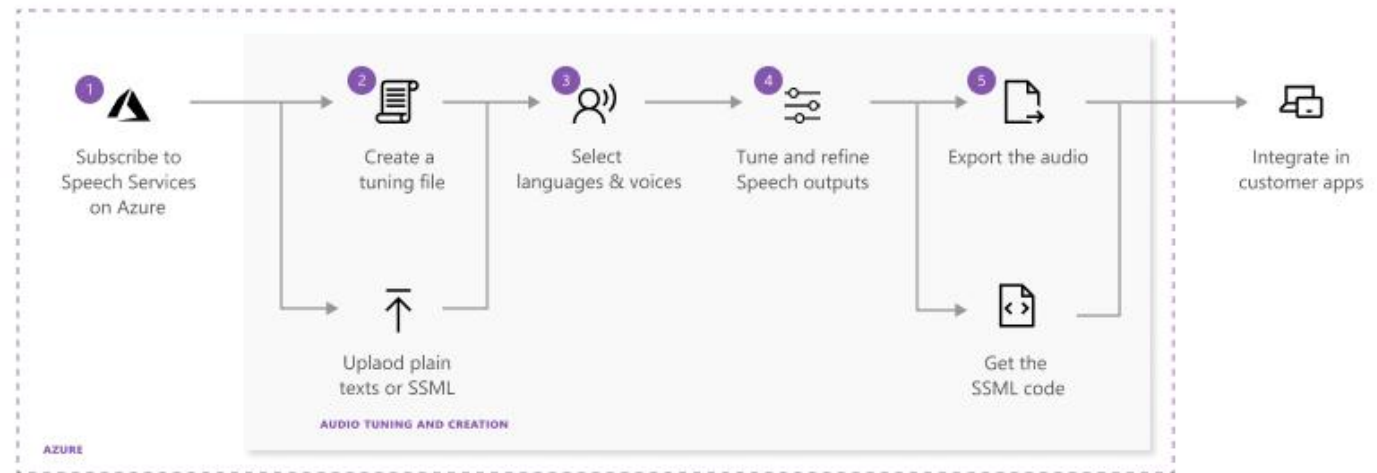
- Build highly natural audio content by fine-tuning text to speech voices and design customized audio experiences.
- Based on Speech Synthesis Markup Language (SSML).
- Adjust text to speech output attributes such as voice characters, styles, speaking speed, pronunciation and prosody.

Good for?

- Audiobooks
- New broadcasts
- Video narrations
- Chatbots

Access?

- Speech SDK
- Speech CLI
- AI Foundry Portal

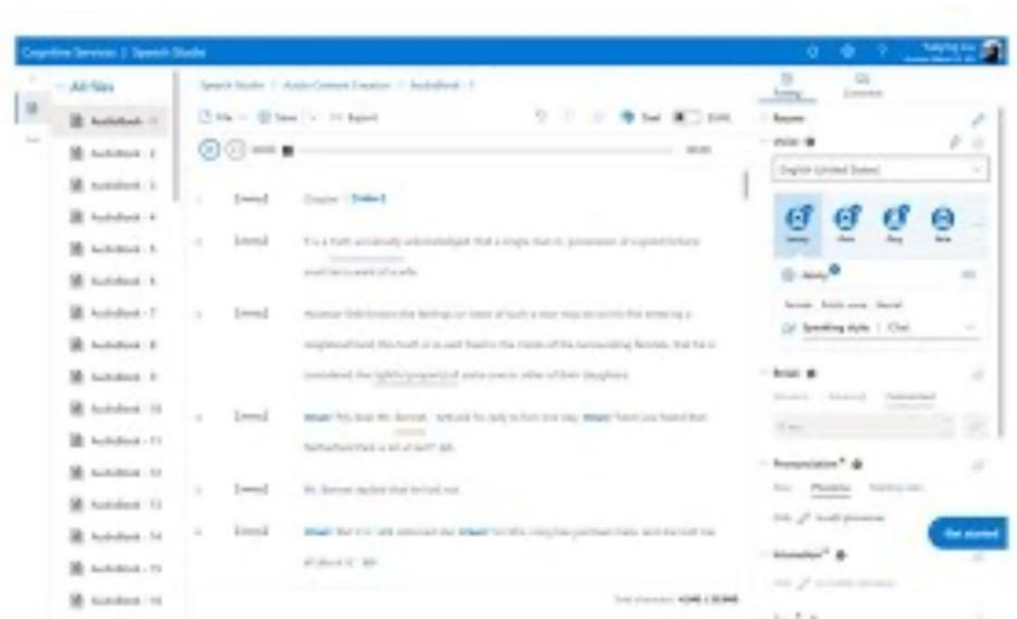


Audio Content Creation

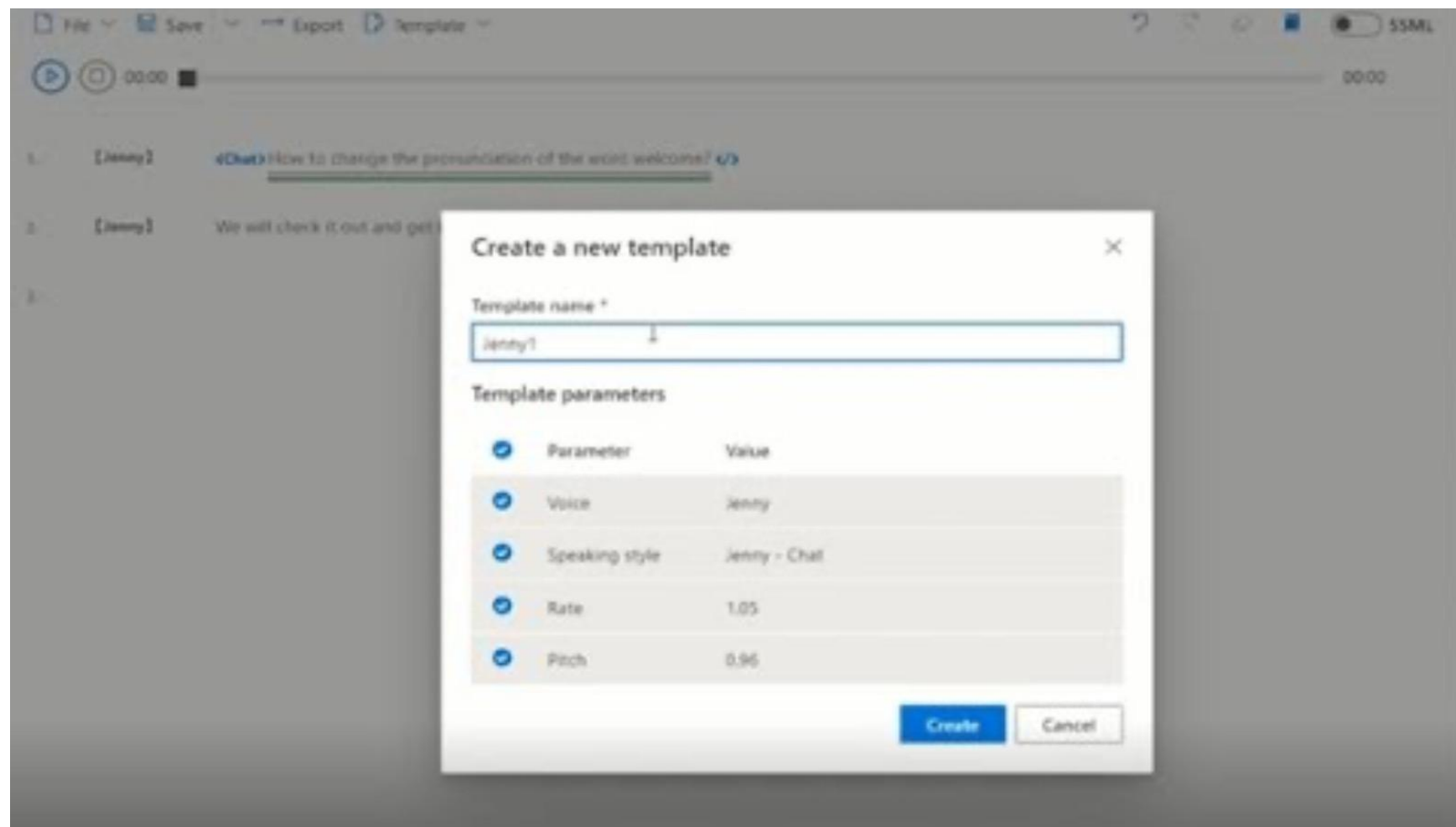


Audio Content Creation

A powerful Speech service tool to visually control AI voices in real-time



New Template



Speech Capabilities by Scenario

Speech Capabilities By Scenario Overview

Captioning with STT

- Convert the audio content of TV broadcast, webcast, film, video or live event into text

Language Learning

- Get instant feedback on pronunciation accuracy, fluency, prosody, grammar and vocabulary

Content Understanding

- New Generative AI designed to process/ingest content of any types such as audio, videos, images

Video Translation

- Translate and apply AI voice dubbing to your videos across 100 languages, with a choice over 400 pre-built voices or using the personal voice

Speech Capabilities Captioning

Captioning

How it works?

- Converting the audio content of a television broadcast, webcast, film, video, live event, or other production into text, and then displaying the text on a screen, monitor, or other visual display system.
- Support real-time and batch.

Good for?

- Online courses and instructional videos
- Sporting events
- Voice and video calls

Access?

- Speech SDK
- Speech CLI
- AI Foundry Portal

Captioning with Speech to Text

[Speech Studio](#) > Captioning

Captioning with speech to text

Use our sample application to learn how to use Azure Speech to automatically caption your content in real-time and offline by transcribing the audio of films, videos, live events, and more. Display the resulting text on a screen to provide an accessible experience. In this example, we leverage features like speech to text and phrase list.

Common use cases:

- Captioning for video content such as films, live television, sports matches
- Transcribing audio-only content like podcasts or phone conversations

[View documentation](#) [Get samples on GitHub](#)

[Try it out](#) [Developer resources](#) [Next steps](#)

Try it out

Choose a video clip to see captioning result.

[Sample videos](#) [Try with your own video](#)

Choose a sample video clip to see real-time or offline processed captioning result.



Real-time captioning



Offline captioning

Technologies used (Speech SDK)

[Speech to text](#)
[Phrase list](#)

Speech Capabilities Language Learning



Clear chat



Stop responding



Download chat audio

Avatar



Off



Hi, What would you like to talk about today?



Press the mic to start talking

For accurate vocabulary and grammar scores, speak at
least 3 sentences before assessment



Clear chat



Stop responding



Download chat audio

Avatar



Off



Hi, What would you like to talk about today?



Press the mic to start talking

For accurate vocabulary and grammar scores, speak at
least 3 sentences before assessment



Hi, What would you like to talk about today?

Let's talk about online shopping.

BZ



Sure! What aspect of online shopping interests you most? For example, benefits, security, or choosing the best platforms?

I would like to talk about the benefit of online shopping.

BZ



Online shopping offers convenience, a broader selection, often better prices, and the ability to compare products and reviews from the comfort of home.

Assess my response



 Recording... stop talking to send

For accurate vocabulary and grammar scores, speak at
least 3 sentences before assessment

Pronunciation score ⓘ



■ 0 ~ 59 ■ 60 ~ 79 ■ 80 ~ 100

Score breakdown

Accuracy Score ⓘ 87 / 100



Fluency Score ⓘ 92 / 100



Prosody Score ⓘ 83 / 100



Content score ⓘ



■ 0 ~ 59 ■ 60 ~ 79 ■ 80 ~ 100

Score breakdown

Grammar Score ⓘ 56 / 100



Vocabulary Score ⓘ 56 / 100



Speech Capabilities

Video Translation

Video Translation

How it works?

- E2E solution for high-quality video translation
- Translation from language A to B and large language model (LLM) reformulation.
- Keep the original timbre, emotions, intonation & style intact
- Subtitles generation
- Multiple-speakers

Good for?

- News + Interviews
- Advertisement + Marketing
- Film + TV Show
- Vlog + Short Video
- Enterprise training

Access?

- REST API
- AI Foundry Portal

aka.ms/videotranslationapi



Source video

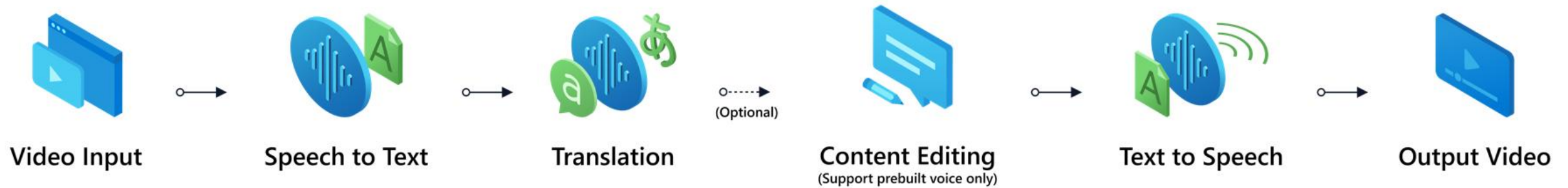


Through Speech API



Translated video

Video Translation Workflow



Video Translation





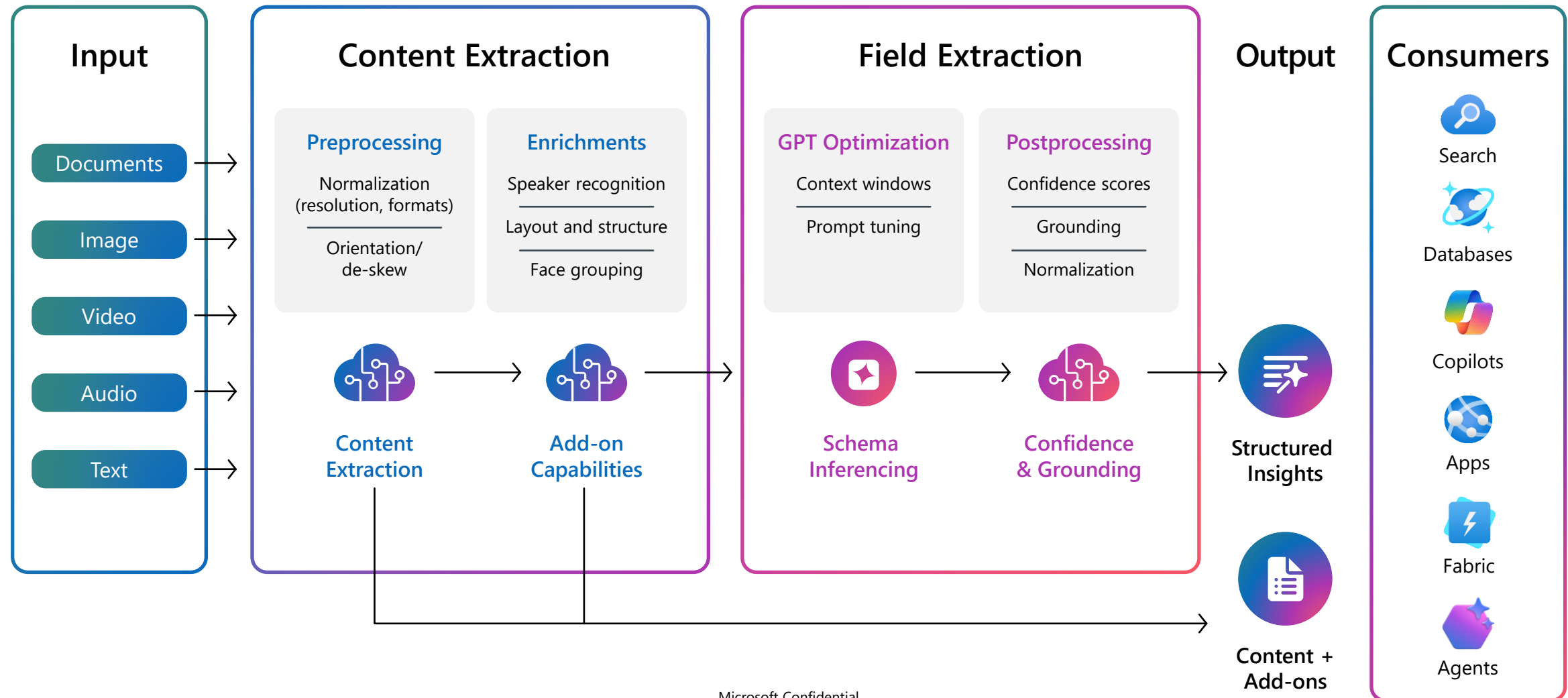
Speech Capabilities

Content Understanding

Azure AI Content Understanding

Built on the success of Document Intelligence, Azure AI Content Understanding offers a new way to reason over large amounts of unstructured data to build customizable workflows, ultimately accelerating time-to-value (TTV), while using a variety of AI models. Ingest and extract any data type into customizable output formats and use your domain expertise to ensure accuracy and improve input.

Components of Content Understanding

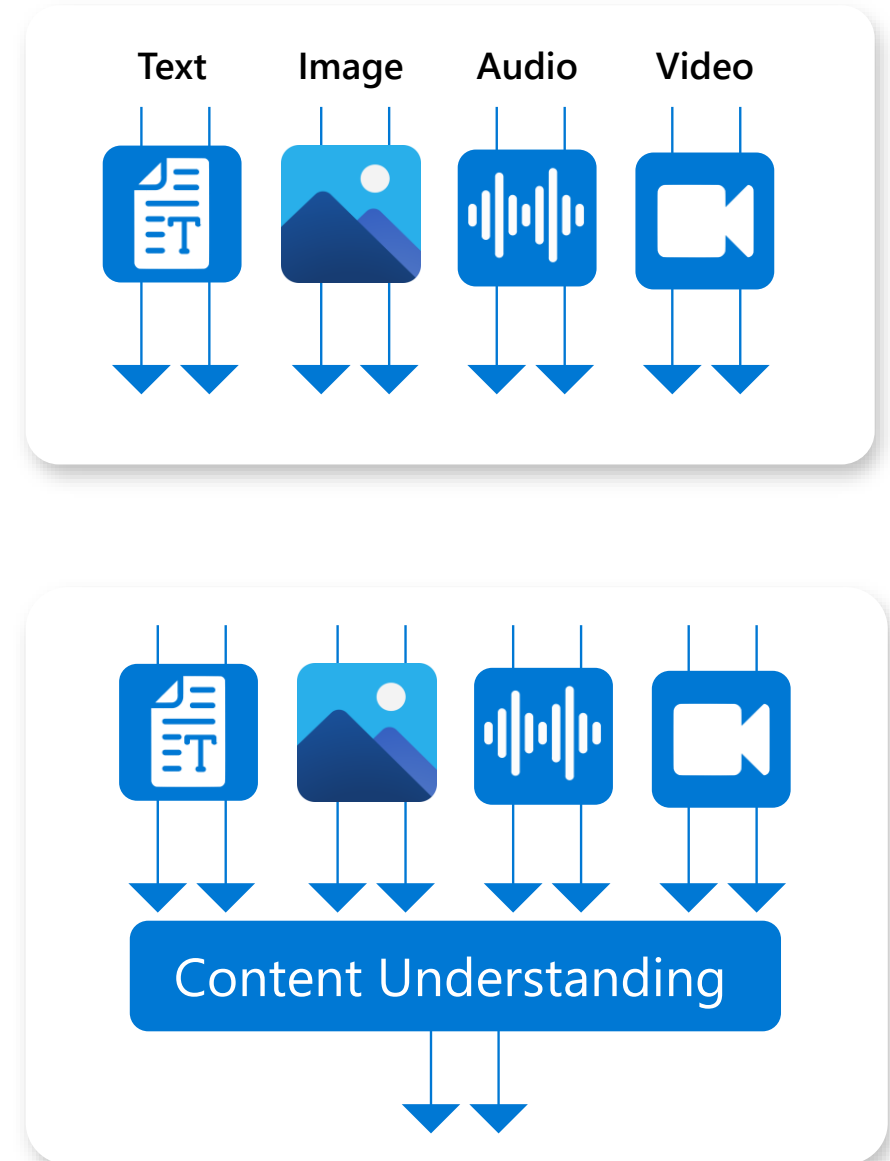


Azure AI Content Understanding (CU)

A new Azure AI service that simultaneously processes multimodal data for the next generation of AI use cases.

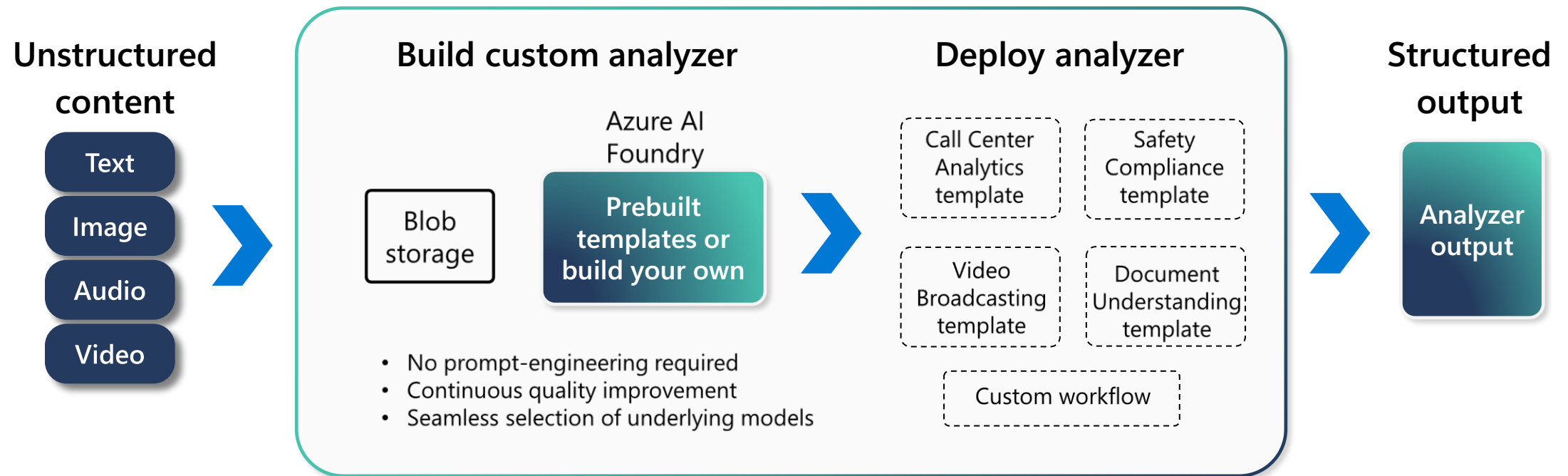
Streamlines the workflow of extracting data from mixed-modal, unstructured content.

Extracts any data type and outputs into customizable structured output formats using a language model (LLM/SLM).



Content Understanding Overview

Azure AI Content Understanding



Azure AI Content Understanding capabilities



Multimodal data ingestion

- Ingest a range of modalities, including documents, images, audio, or video.
- Use a range of AI models in Azure AI to convert input data into a structured format.
- Process and analyze through downstream services or applications.



Information extraction schema

- Define schemas of extracted results.
- Generate task-specific representations of the output, including insights, features, or summaries.
- Build enterprise GenAI apps or agentic workflows to automate business processes through LLMs and RAG.



Grounding and confidence scores

- Ensure schema-identified values are accurate and usable with GenAI tools.
- Ground extracted information in underlying content.
- Provide confidence scores to reduce human intervention.
- Enable continuous improvement through user feedback.

Customers are adopting multimodality across industries



Health Care

- Generating and completing medical reports
- Improving patient interactions



Customer Service

- Call center analytics
- Automated call summaries
- Chatbot integration



Document Processing

- Contract analysis
- Claims processing



Common across industries

- Knowledge management
- Compliance recording

Use Cases



Video analysis and broadcasting

Audio + Video

- Generate task-specific representations, including insights, features, and summaries.
- Automatically generate chapters and scenes within videos for easier editing, remixing.
- Insert contextual ads based on video and audio being extracted and analyzed.



Document understanding

Document + Images

- Build information lookups or RAG scenarios with extracted content.
- Extract information from documents that include text, tables, and images of charts.
- Analyze extracted information using foundational models.
- Enable translations in other languages for smoother communication.

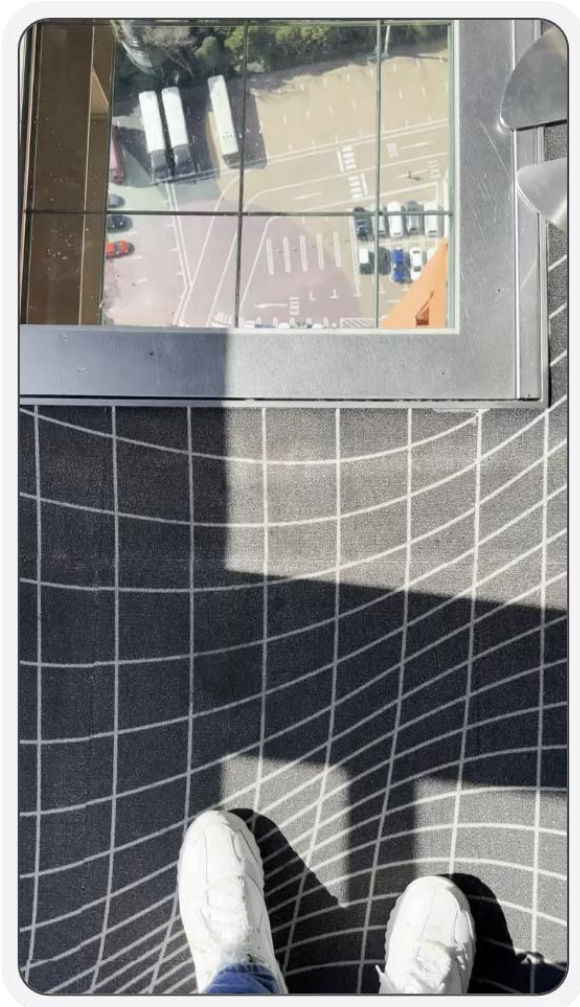


Post-call analytics

Audio + Video + Document

- Ingest a variety of content, such as audio or video call recordings.
- Transcribe, analyze, and summarize the call.
- Ensure compliance across call center agents and customer-employee conversations.

Social Media Trends



Description

"The video appears to be a travel vlog showcasing various experiences and scenes in [Tokyo, Japan](#). It features views from a high-rise building, scenic shots of a historic bridge and moat, and various scenes of daily life and public transport. Notably, the video highlights the [unique vending machines](#) found in Tokyo, offering items from sandwiches and umbrellas to more unique items like cups of noodles and flowers. People are seen engaging in everyday activities such as walking through streets, using public transport, and interacting with vending machines. The video captures the essence of Tokyo's urban life and its blend of traditional and modern elements."

Products

- "Sandwich Vending Machine"
- "Flower Vending Machine"
- "Umbrella Vending Machine"
- "Cup Noodles Vending Machine"
- "Custard Sandwich"
- "Fragrance Rose Cup"

Locations

- "High-rise building overlooking a cityscape"
- "Historic bridge and moat"
- "Urban streets"
- "Subway station"
- "Various vending machine locations"

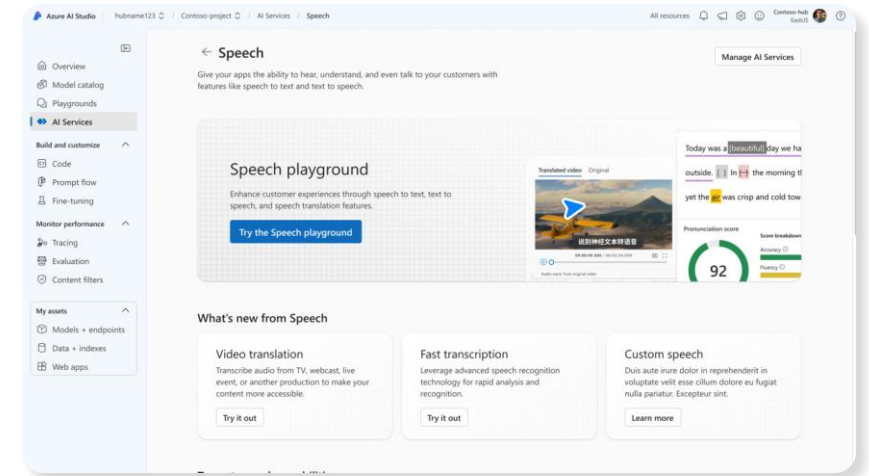
Resources

	Announcement Blog	aka.ms/content-understanding-launch-blog
	Breakout session @ Microsoft Ignite 2024	youtu.be/vYjAg27aHkA?si=k73mzRiA7tgEOhoz
	Product Page	aka.ms/content-understanding
	Pricing Page	azure.microsoft.com/pricing/details/content-understanding
	Docs	aka.ms/content-understanding-mslearn
	AI Studio	aka.ms/content-understanding-aistudio
	Samples Repositories	aka.ms/cu-samples github.com/Azure-Samples/azure-ai-content-understanding-python github.com/Azure-Samples/azure-ai-search-with-content-understanding-python github.com/Azure-Samples/azure-ai-content-understanding-with-azure-openai-python

Enhanced developer experience

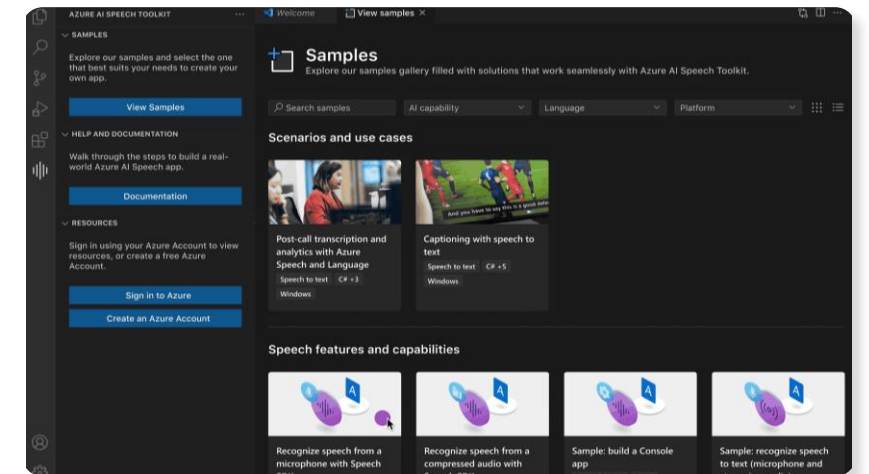
- Speech playground in AI Foundry
- Playgrounds
- Custom and fine-tuning experiences
- Everything together in one place

ai.azure.com/explore/aishervices/speech



- Toolkit extension in Visual Studio code
- Gallery of speech-enabled **quick-starts** and **scenario samples**;
- Seamlessly **create** and **manage** azure resources in place;
- Easily **build** and **run** code samples with clicks.

aka.ms/speech-toolkit-vscode



Questions?